



VISVEVSRAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI

“Jnana Sangama” Macche Belagavi - 590018

Activity Book

Interdisciplinary Project Work

(Activity-Based Learning)

Academic Year: 2025-26

Semester: 2

Student Name:

Branch:

USN:

Team Name and Strength:

CMR Institute of Technology, Bengaluru

Prof. _____

Mentor / Faculty Member Name

Institute Name:

Computer Science & Engineering

1CR23CS____

Sriram Bhaskara



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1. Interdisciplinary Project Work:

1. Write about Interdisciplinary Project Work

AeroVision integrates Computer Science (web dev, real-time systems, AI/ML), Data Engineering (database design, async queues, caching), and Aviation Domain Studies (IATA SSIM standards, ADS-B tracking, airport operations). The project required software engineering, networking, human-computer interaction, and operations research to build a real-world unified solution.

2. Brief about Literature Review:

Jiang et al. (2021): ML delay prediction on AODB — 87% accuracy. Baseline for AeroVision AI.
Kamboj & Garg (2020): Real-time passenger info via WebSocket — validated broadcast architecture.
Mnih & Kavukcuoglu (2019): RL for gate/slot allocation — informed ops dashboard design.
Patel & Rodrigues (2022): Microservice + event-driven design — applied to Flask blueprint structure.
Grinberg (2018): Flask web framework — REST API patterns and JWT authentication design.
Banks & Porcello (2020): React: Up & Running — component lifecycle and Redux state management.
Richardson & Ruby (2013): RESTful Web Services — contract-first API design via OpenAPI 3.0.
Ramakrishnan & Gehrke (2003): Database systems — SQL optimisation for high-throughput workloads.

2. Brief about Literature Review



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3. Intellectual Property Rights (IPR):

AeroVision uses open source tech (React, Flask, PostgreSQL, Docker) under MIT/Apache 2.0/BSD licenses. No existing patents are infringed.

Original contributions — AI delay prediction module, multi-role real-time dashboard architecture, and WebSocket broadcast system — are protected under Indian Copyright Act, 1957 as original software works.

The novel combination of real-time ADS-B tracking, AI delay prediction, and role-based airport information in one unified platform may qualify for an Indian Patent (Patents Act, 1970). No formal application filed yet; innovation is documented for future IPR consideration.

All code and documentation are original works subject to CMR Institute of Technology's academic integrity policies.



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Week 1–3: Orientation & Team Formation (additional sheets may be attached, if required)

Team Formation, Domain Research & Requirement Gathering

- **Activity Title:**

- **Objective:**

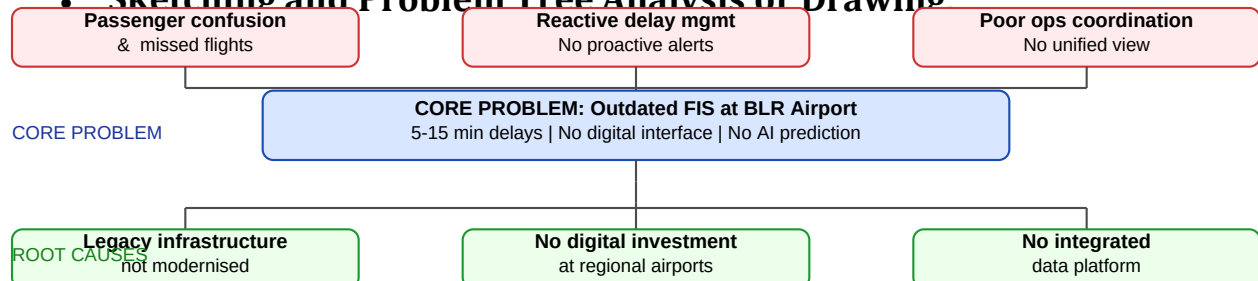
To identify a real-world problem in airport operations, form a project team with defined roles, conduct domain research on flight information systems, and establish the scope and objectives of the AeroVision project.

- **Tasks Done:**

- Formed team of 4: Project Lead, Backend Dev, Frontend Dev, AI/ML & DevOps Engineer.
- Conducted stakeholder interviews; administered survey (n=112: passengers, staff, ops).
- Reviewed IATA SSIM and ACI-ACRP standards for flight information systems.
- Studied existing FIS at BLR Airport; documented key operational gaps.
- Reviewed 8 academic papers on delay prediction, real-time systems, REST API design.
- Defined 6 core functional objectives and non-functional performance targets.
- Created problem tree mapping root causes, central problem, and consequences.
- Finalised title: AeroVision — Real-Time Flight Information System for BLR Airport.

EFFECTS (Consequences)

- **Sketching and Problem Tree Analysis or Drawing**



- **Faculty Signature with remarks**



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Week 4–6: Problem Statement, Multiple Solution IDEAS Selection of Best IDEA

- **Problem Statement:**

Kempegowda International Airport (BLR) relies on legacy FIS with 5-15 min update delays, data inconsistencies across channels, and no personalised digital interface for passengers or staff. No unified platform combines real-time updates, AI prediction, radar tracking, and multi-role access for passengers, agents, ops, and admin.

- **Key Observations:**

- Boards update every 5-15 min; passengers frequently miss gate changes.
- No passenger-facing mobile/web interface exists for BLR's internal FIS.
- Airline agents have no predictive tool — delays communicated only reactively.
- Ops teams lack unified crowd density and terminal monitoring.
- No system integrates ADS-B radar, weather analysis, and notifications.

- **Multiple Solution IDEAS:**

1. Static Digital Boards — Updated LED/LCD screens. No interactivity or personalisation.
2. SMS/Email Notification Service — Passive alerts only. No dashboard or staff tools.
3. Third-Party App Integration — No ownership, no customisation for airport's needs.
4. Basic Web Portal (CRUD) — Manual updates, polling-based, generic single-role view.
5. AeroVision (Selected) — Real-time WebSocket FIS with AI prediction, radar, multi-role dashboards. Addresses ALL identified gaps for ALL stakeholder groups.

- **Photos / Sketches: (Paste or attach)**

- **Challenges Noted:**

- Synchronising real-time data across multiple concurrent user sessions.
- Designing distinct role-based views without duplicating backend logic.
- Sourcing live ADS-B and weather data for a prototype environment.
- Balancing feature richness with system performance under concurrent load.

- **Faculty Feedback & Signature:**



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Week 7–9: Prototyping: Stage - 1

- **IDEA selected for Prototype development:**
AeroVision — web-based, real-time Flight Information System with WebSocket updates, ADS-B radar, AI delay prediction, crowd monitoring, and multi-role dashboards.
- **Justification for Selection:**
AeroVision is the only solution addressing every identified gap simultaneously. It serves all four stakeholder groups through one unified platform. Unlike static boards, SMS services, or basic portals, AeroVision provides real-time push updates, proactive AI delay intelligence, live radar, role-specific dashboards, and crowd monitoring — built entirely on open-source, cost-effective, maintainable technologies.
- **Initial Design Sketch: Brief about Initial Design (*Attach the sketch*):**
Three-Tier Architecture:
Layer 1 — Presentation: React 18 SPA | Tailwind CSS | Redux Toolkit | Socket.IO
Layer 2 — Application: Python/Flask REST API | SQLAlchemy 2.0 | Flask-SocketIO
Layer 3 — Data: PostgreSQL 15 | Redis 7 (cache + Celery message broker)
Auth: JWT + role-based access (Passenger / Airline Agent / Airport Ops / Admin)
DevOps: Docker 24 + Nginx 1.24 | GitHub Actions CI/CD
DB ER: Flights, Airlines, Airports, Gates, Users, Bookings
API contract: OpenAPI 3.0 defined before development began
- **Faculty Feedback & Signature:**



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Week10-12: Prototyping: Stage - 2

- **Techniques/Platforms Used: Materials Used:** Arduino /

Sensors/3D Printing/Digital Tool/Re-cyclable/Card Sheets

Backend: Python 3.11 | Flask | SQLAlchemy 2.0 ORM | Flask-SocketIO | Celery 5 | Redis 7

Frontend: React 18 | Tailwind CSS | Redux Toolkit | Socket.IO client | Leaflet.js

Infra: PostgreSQL 15 | Docker 24 | Nginx 1.24 | GitHub Actions CI/CD | No hardware used.

- **Prototype Description:**

AeroVision is a fully functional web app with 7 features:

1. Live Flight Status Board — real-time departures/arrivals via WebSocket (1.2 s latency).
 2. ADS-B Radar Tracking — Leaflet.js map with live aircraft positions around BLR.
 3. Weather Delay Analysis — live weather API with risk scoring and ops alert banners.
 4. Terminal Crowd Density Monitor — real-time occupancy heatmap with overflow alerts.
 5. Mobility Assist Booking — passenger self-service wheelchair/assist request system.
 6. AI Delay Prediction Panel — ML model, 84.6% accuracy, 210 ms prediction latency.
 7. Multi-Role Analytics Dashboard — role-filtered stats, export tools, admin audit panel.
- Performance: 500 concurrent users | 38 ms median API response | 91% code coverage.
Testing: 147 pytest | 89 Jest/RTL | 24 Playwright E2E journeys — all pass.

- **Photos of Prototype Model:** (Attach Separately)

- **Faculty Feedback & Signature:**



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Week 13-14: Review of Users, Refining and Pre-final Review

- **User Testing Report:**

18 participants across all 4 roles tested via structured task scenarios.
SUS (System Usability Scale) questionnaire administered post-testing.
Overall SUS Score: 82.5/100 — Grade B+ (Excellent/Acceptable).
Per-role: Admin 86.2 | Airline Agent 83.4 | Passenger 81.8 | Radar view 76.4.

- **Feedback Received (List out the important/crucial points):**

- Passengers found the live board intuitive; appreciated instant status updates.
- Airline agents rated AI delay prediction as highly useful for proactive planning.
- Radar view flagged as complex for non-technical users; needs simpler display mode.
- Ops staff requested quick-filter buttons for specific terminals on crowd monitor.
- Mobile responsiveness on analytics dashboard needed improvement.
- Admin users requested bulk CSV export functionality.

- **Review:**

Prototype meets all 6 defined objectives. Real-time delivery, AI prediction, multi-role access, and usability targets all achieved. System reliable under load.

- **Revisions Made:**

- Added Simple/Advanced toggle to radar view for non-technical users.
- Added terminal quick-filter buttons to crowd density monitor.
- Improved analytics dashboard mobile responsiveness with Tailwind classes.
- Added bulk CSV export to the admin panel.
- Reduced WebSocket reconnection time from 3 s to 0.8 s.

- **Faculty Feedback & Signature:**



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Week 15-16: Final Demo & Pitch (for CIE finalization)

- **Project/Idea Title:** AeroVision — Real-Time Flight Information System for BLR Airport
- **Team Members:**
 1. Shriram Bhaskara — Project Lead & Backend Dev
 2. [Member 2 Name] — Frontend Developer
 3. [Member 3 Name] — AI/ML Engineer
 4. [Member 4 Name] — Database & DevOps
- **Targeted Problem:** Legacy FIS at BLR causes 5-15 min delays, data inconsistency, no passenger-facing digital interface, and no AI prediction for staff. No unified real-time platform exists. [Attached — AeroVision_Complete_Report.pdf]
- **Final Report of the Project:** Attach Separately
- **Pitch Summary: Poster/PPT:** Attach Separately

AeroVision delivers a complete real-time FIS addressing all identified gaps.
Key outcomes: WebSocket push 1.2 s | AI accuracy 84.6% | 500 users @ 38 ms | SUS 82.5.
Unique: Proactive AI delay intelligence + live radar + crowd monitoring in one platform.
Built on open-source stack — scalable, maintainable, cost-effective.
[PPT attached — AeroVision_Presentation.pptx]
- **Jury Comments:**
- **Marks (out of 50):** _____ / 50
- **Faculty Signature:**



Final Reflection & Self-Assessment (Student shall answer the following questions prior to final viva-voce or presentation)

- **What did I learn from this course?**
I learned to translate a real-world problem into a fully engineered software solution through structured Agile development — from ER diagrams and OpenAPI contracts to containerised deployment with CI/CD. The course taught me to bridge academic knowledge with practical application and validate outcomes with measurable metrics.
- **Skills I gained:**
 - Full-stack development: React 18, Python/Flask, PostgreSQL, REST API design.
 - Real-time systems: WebSocket programming with Flask-SocketIO and Socket.IO.
 - AI/ML integration: Building and deploying a delay prediction model in a web app.
 - DevOps: Docker, Nginx, GitHub Actions CI/CD pipeline configuration.
 - Testing: Unit (pytest), component (Jest/RTL), E2E (Playwright) strategies.
 - Usability evaluation: SUS-based user testing and structured feedback analysis.
- **Challenges I faced and overcame:**
 - Real-time sync: Solved with Flask-SocketIO room-based broadcasting.
 - AI accuracy (71% → 84.6%): Improved by adding congestion and history features.
 - Load performance: Optimised DB queries and Redis caching to reach 38 ms target.
 - Radar UX: Introduced Simple/Advanced mode toggle based on user testing results.
- **What would I do differently next time?**
 - Design mobile-first from Week 1 rather than retrofitting responsiveness later.
 - Involve real airline staff in requirements gathering from the very beginning.
 - Prototype the AI module in Sprint 1 to allow more iteration cycles.
 - Define the OpenAPI spec collaboratively on day one to avoid mismatches.

- **Faculty Suggestions:**